

2017

Reference Books:

1. Computer Security by D. Gollmann, 3rd edition, 2011
2. Computer Security Fundamentals by C. Easttom, 4th edition, 2019
3. The Official (ISC)2 Guide to the CISSP CBK Reference by J. Warsinske, 1st edition, 2019
4. Information Security: Principles and Practice by M. Stamp, 2nd Edition, 2011

WEEKLY LIST OF EXPERIMENTS

Sr.	Topics	LLO, GA
1	Introduction to Security Terms and Concepts Understanding Security Attacks Exploring Security Mechanisms	LLO1, GA2
2	The Need for Security ,Identifying Business Needs for Security ,Analyzing Simulating Common Attacks, Different Threats	LLO3, GA5
3	Implementing Classical Encryption Techniques Encryption and Decryption with Symmetric Ciphers Exploring Block Cipher Principles	LLO2, GA4
4	Understanding Transposition Ciphers Implementing Rail Fence Cipher Analyzing Feistel Cipher	LLO2, GA4
5	Hands-on with DES Encryption, Implementing Advanced Encryption Standards (AES)	LLO3, GA5
6	Implementation RSA Algorithm, Key Exchange with Diffie-Hellman Exploring Hash Functions	LLO2, GA4
7	Case Studies on Legal and Ethical Dilemmas	LLO3, GA5
8	Lab Sessional	LLO1, GA2
9	Developing Risk Control Strategies Simulation Risk Management	LLO2, GA4
10	Developing Information Security Policies Creating Security Education Programs Blueprint Analysis (ISO-27000, NIST, IETF)	LLO3, GA5
11	Configuring Access Control Policies, Setting up Firewalls, VPN Implementing Intrusion Detection Systems Setting up Honeypots and Honeynets Testing Scanning and Analysis Tools	LLO3, GA5
12	Open Ended Lab	LLO3, GA5
13	Designing Physical Access Control Systems Fire Security and Safety Mobile Designing Portable Systems Security Testing	LLO2, GA4
14	Simulation Information Security Project Management Implementation Technical Challenges Implementation Non-technical Strategies	LLO2, GA4

15	Implementation Asymmetric encryption (RSA), hash functions (MD5, SHA-256), and digital signatures	LLO2, GA4
16	Lab Sessional	LLO1, GA2